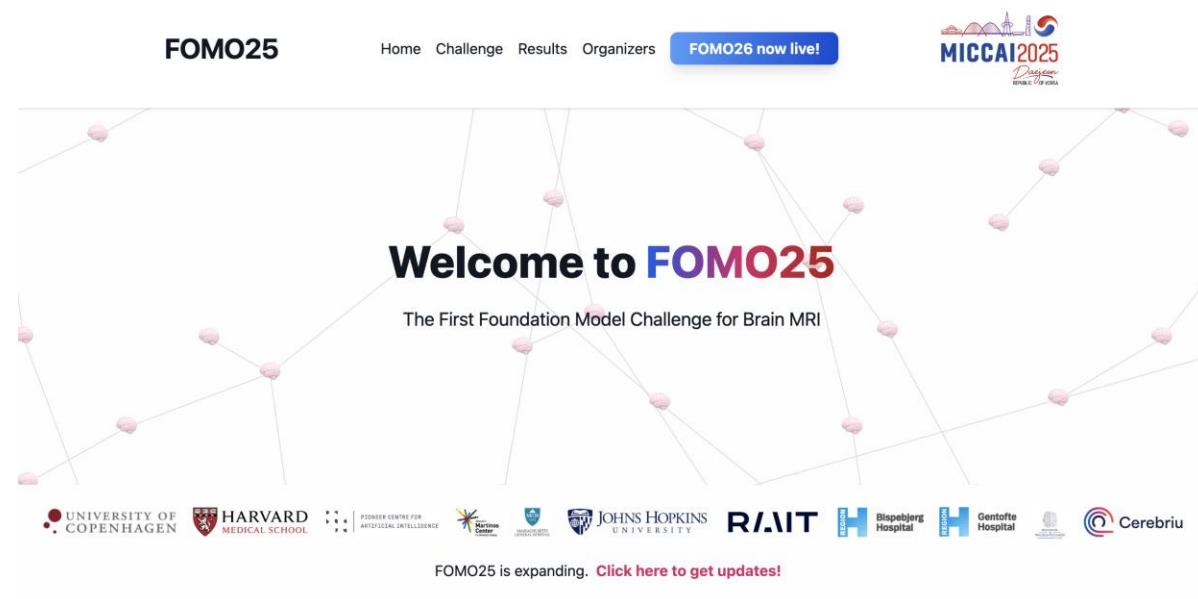


Towards Brain MRI Foundation Models for the Clinic: Findings from the FOMO25 Challenge



Department of Computer Engineering and Artificial Intelligence
MAI Lab.

YS Park

- Introduction
- Data
- Results
- Discussion & Limitations & Conclusion
- FOMO26

- **Neuroimaging machine learning의 dominant paradigm**
 - **task-specific supervised model**을 manually annotated data로 학습
 - 이 방식은 internal validation set과 public benchmark에서 강력한 성능을 보였지만, Domain shift 취약성과 Structural asymmetry (소량만 label됨)이라는 근본적 한계를 가짐
- **SSL이 해결책인 이유**
 - 각 task별 separate predictor 학습 대신, **large unlabeled corpus로 encoder를 먼저 학습한** 뒤 specific task로 adaptation
 - NLP와 computer vision에서 foundation model이 **label-efficient, robust to distribution shift, cheap to finetune**임이 입증됨

- **기존 Brain MRI Foundation Model의 한계**
 - Small pretraining dataset (<10K scan)
 - Narrow evaluation: Clean, research-grade downstream benchmark에 국한
 - 이런 setting에서 supervised model이 이미 잘 작동하므로, foundation model은 cosmetic improvement만 평가받는 구조
- **FOMO25**
 - **FOMO60K** 공개: 60,529개 structural brain MRI scan
 - **Clinical workflow에서 직접 sourcing된 data**로 평가
 - **Out-of-domain generalization + label efficiency track:**
 - Method track (FOMO60K로 제한) + Open track (제약 없음)
 - **downstream task:**
 - infarct classification
 - meningioma segmentation
 - brain age regression

- 참여 규모 및 핵심 Findings

- 137 registration, 30 participant, 18 team, 19 foundation model 평가

- Key Findings (1):

- **Self-supervised pretraining boosts generalization:** SSL이 동일 OOD data로 학습된 supervised baseline을 넘어, in-domain specialist model까지 능가

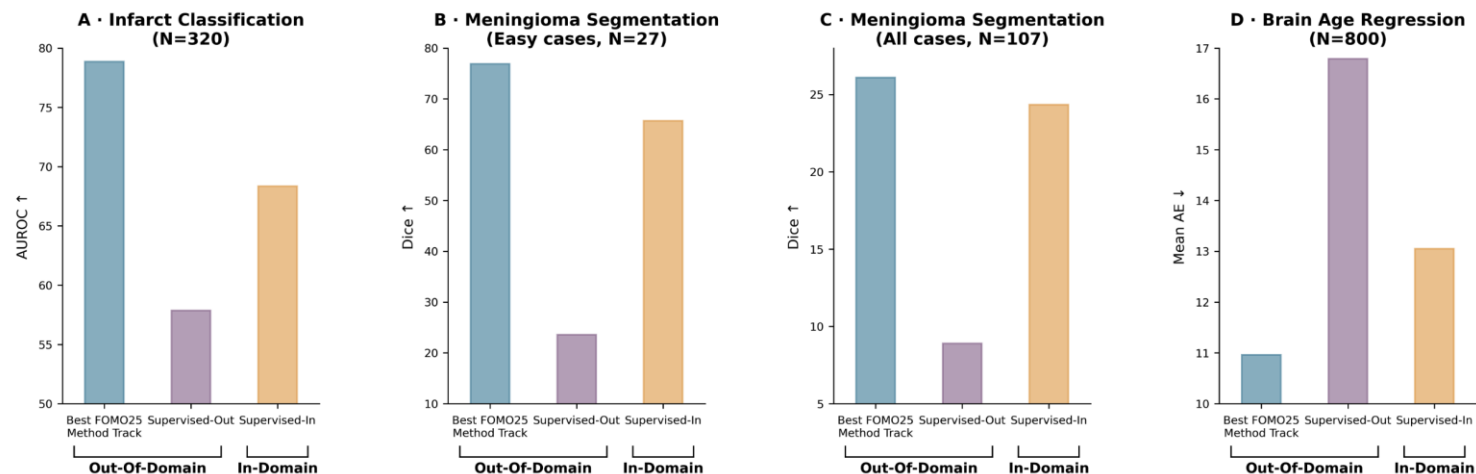


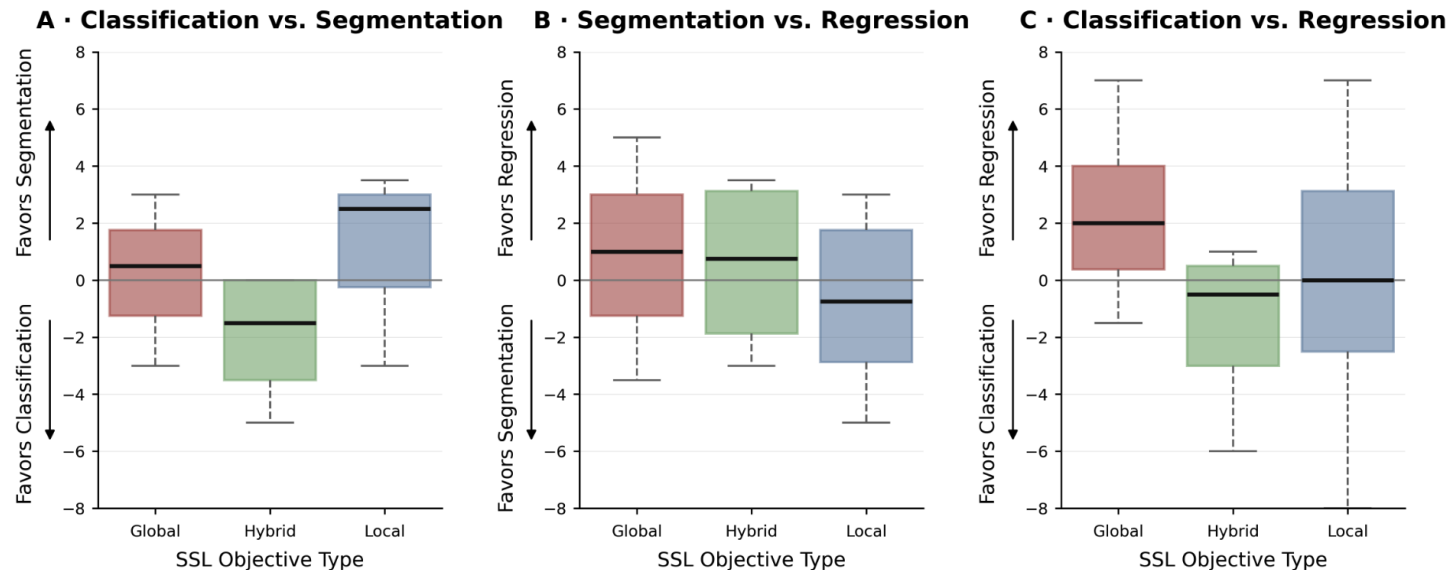
Figure 1: **Self-supervised pretraining boosts generalization.** Across tasks, the top pretraining-based model from the *method-track* outperforms both from-scratch out-of-domain and in-domain supervised baselines, demonstrating that SSL can effectively leverage heterogeneous MRI data. Baselines are nnU-Net (segmentation) and Asparagus (classification/regression). The best performing method for classification was ■ashash, for segmentation ■Dolphin_creators, and for regression ■FOMO2JOMO. For Task 2, “Easy cases” were defined as the top 25% of cases ranked by average Dice score across all challenge participants.

- 참여 규모 및 핵심 Findings

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- Key Findings:

- **No single SSL objective benefits all tasks:** Local objective(MAE)는 segmentation 유리, hybrid(MAE+contrastive)는 classification 유리, regression은 모든 objective가 비슷

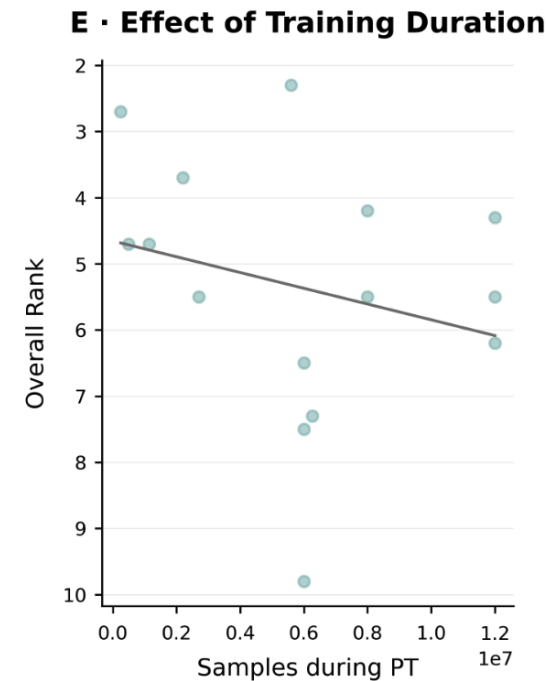
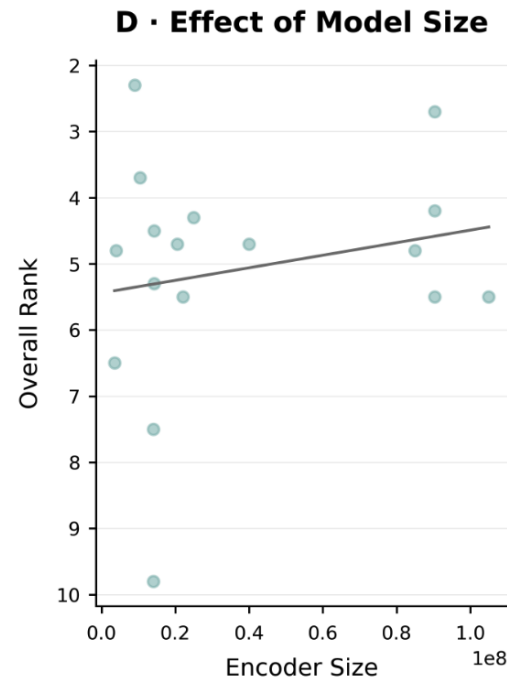


- 참여 규모 및 핵심 Findings

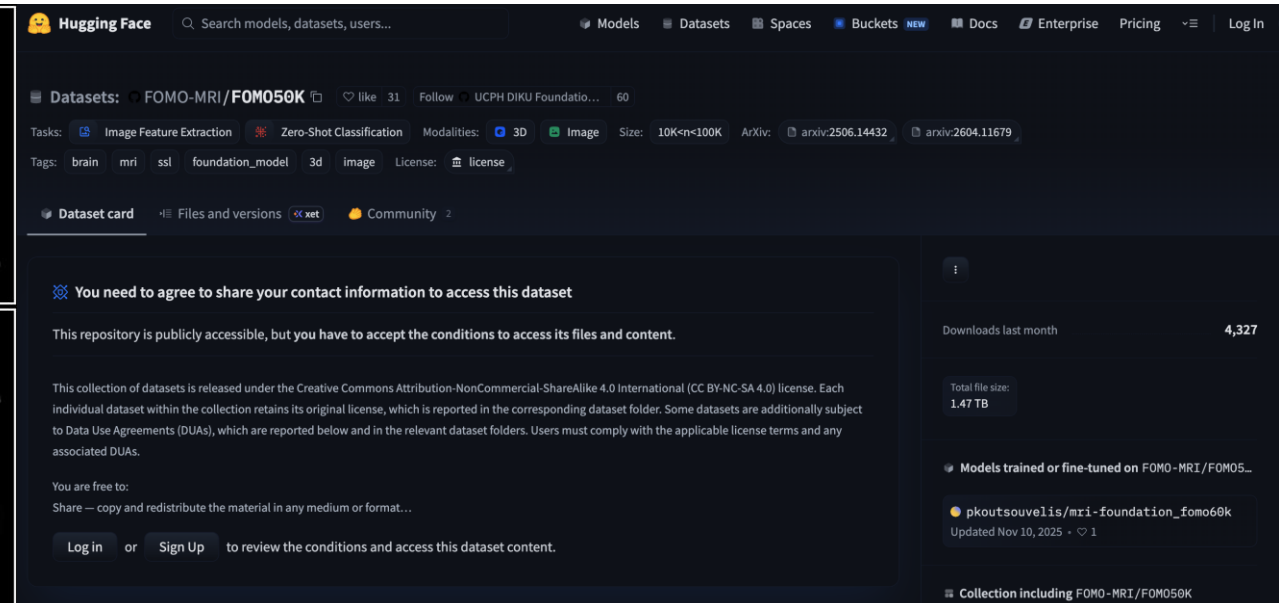
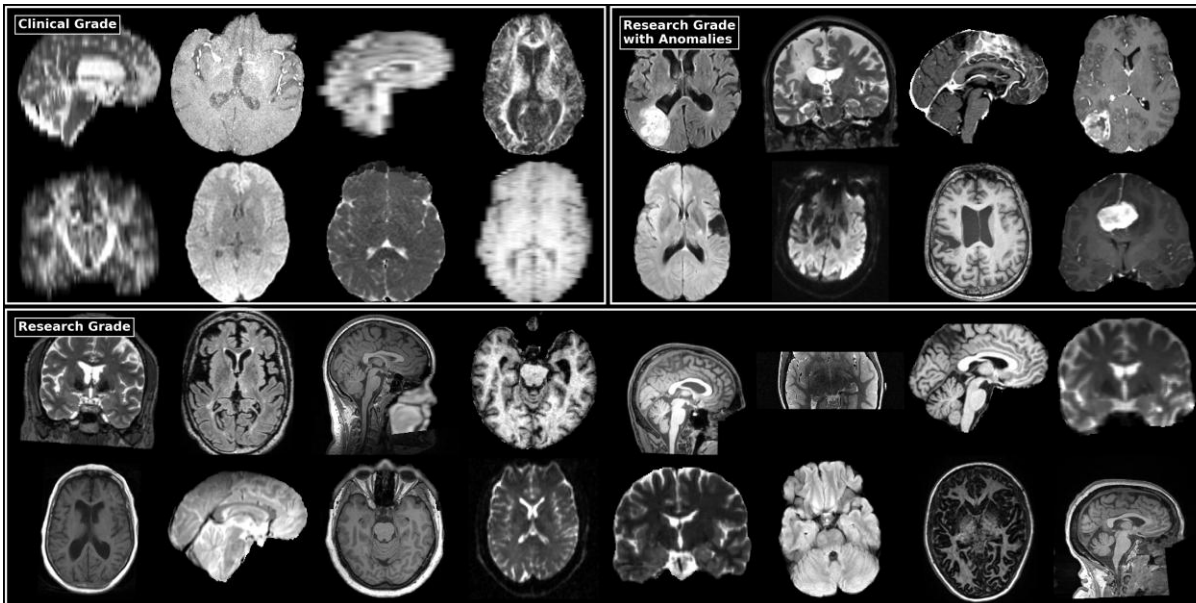
- 137 registration, 30 participant, 18 team, 19 foundation model 평가

- Key Findings:

- **Small models were competitive:** Model size와 pretraining duration의 scaling 효과가 미미함



- Pretraining Data - FOMO60K
 - 60,529 MRI scans / 13,900 sessions / 11,187 subjects
 - 16개 public source에서 하나로 모은 Dataset
 - Heterogeneous: subject demographics, scanner vendor/model, field strength, acquisition protocol, MRI sequence type
 - 전처리: RAS orientation 변환, session 내 highest-resolution scan에 affine co-registration, skull-stripping 또는 defacing



- Finetuning Datasets

Task	Description	Cases	Origin	Sequences
Task 1	Infarct	21 (13 pos / 8 neg)	India (Cerebriu)	FLAIR, DWI (ADC + b1000), T2*/SWI
Task 2	Meningioma	23	India (Cerebriu)	FLAIR, DWI (b1000), T2*/SWI
Task 3	Brain Age	200	USA (Boston)	T1w, T2w

- Validation/Test Datasets — Out-of-Domain

Task	Validation		Test	
Task 1	80	320 (positive/negative balanced)		AUROC
Task 2	25		107	Dice + NSD
Task 3	200		800	MAE + corr.

- Baseline
 - Supervised-OOD: Out-of-domain finetune dataset으로 학습 (참가자와 동일 조건)
 - Supervised-ID: Validation dataset으로 학습 (in-domain, 더 큰 규모)
 - Baseline Models:
 - Classification/Regression: Asparagus framework (ResEnc, 90M params, GAP + linear head, SGD)
 - Segmentation: nnU-Net (default planner, full-dataset training configuration)
- Rankings and Statistical Analysis
 - Task 내 metric-specific rank 평균 → task 간 평균 → overall ranking
 - 10,000 bootstrap resample로 95% CI 산출
 - 100,000 permutation test (Phipson-Smyth correction)로 pairwise significance 평가
 - Per-task p-value의 maximum을 결합 → 한 team이 superior로 판단되려면 모든 task에서 outperform 필요
 - Performance tier ($\alpha=0.05$): 평균 rank로 정렬 후 significant difference 없는 team끼리 grouping

- Rankings and Statistical Analysis

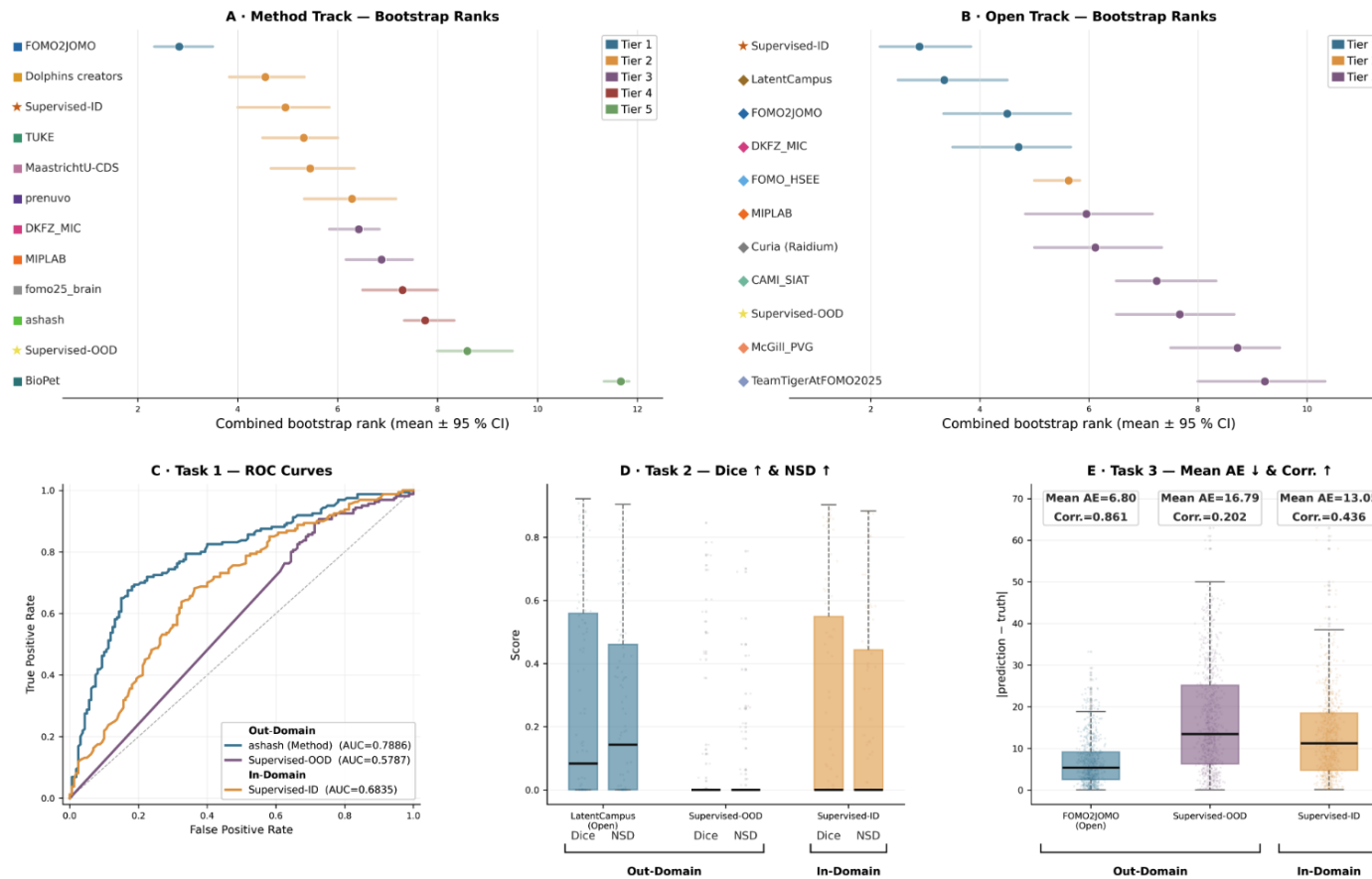


Figure 6: **Rankings with supervised baseline models.** (A, B) Bootstrap rank distributions (mean $\pm$ 95% CI) for the Method and Open tracks, with teams grouped into statistically defined performance tiers (colours). (C-E) Task-level performance for the best out-of-domain method across both tracks alongside in-domain and out-of-domain supervised baselines: (C) ROC curves for the infarct classification task, (D) Dice and NSD score distributions for the meningioma segmentation task, (E) absolute error distributions for the brain-age regression task, with Mean AE and Corr. annotated. Rankings per individual task, along with permutation-test p-value matrices, are provided in Figure A.7 for the Method track and in Figure A.8 for the Open track.

- Choice of pretraining strategy

- Local: Voxel-level supervision (모두 MAE variant)
 - Segmentation > Classification 경향, regression과는 차이 없음
- Global: Sequence/subject-level (contrastive, supervised 등)
 - 의외로 segmentation > classification > regression 경향
- Hybrid: Local + global 결합 (MAE+Contrastive, DINOv2 등)
 - Classification과 regression > Segmentation 경향
- 모든 model이 task 간 rank variation이 큼 → universally good model 없음
- Encoder size vs overall rank: 약한 upward trend만 존재 (correlation 약함)
- Pretraining sample 수 vs rank: 의미 있는 개선 없음 → early saturation

- Effect of augmentation

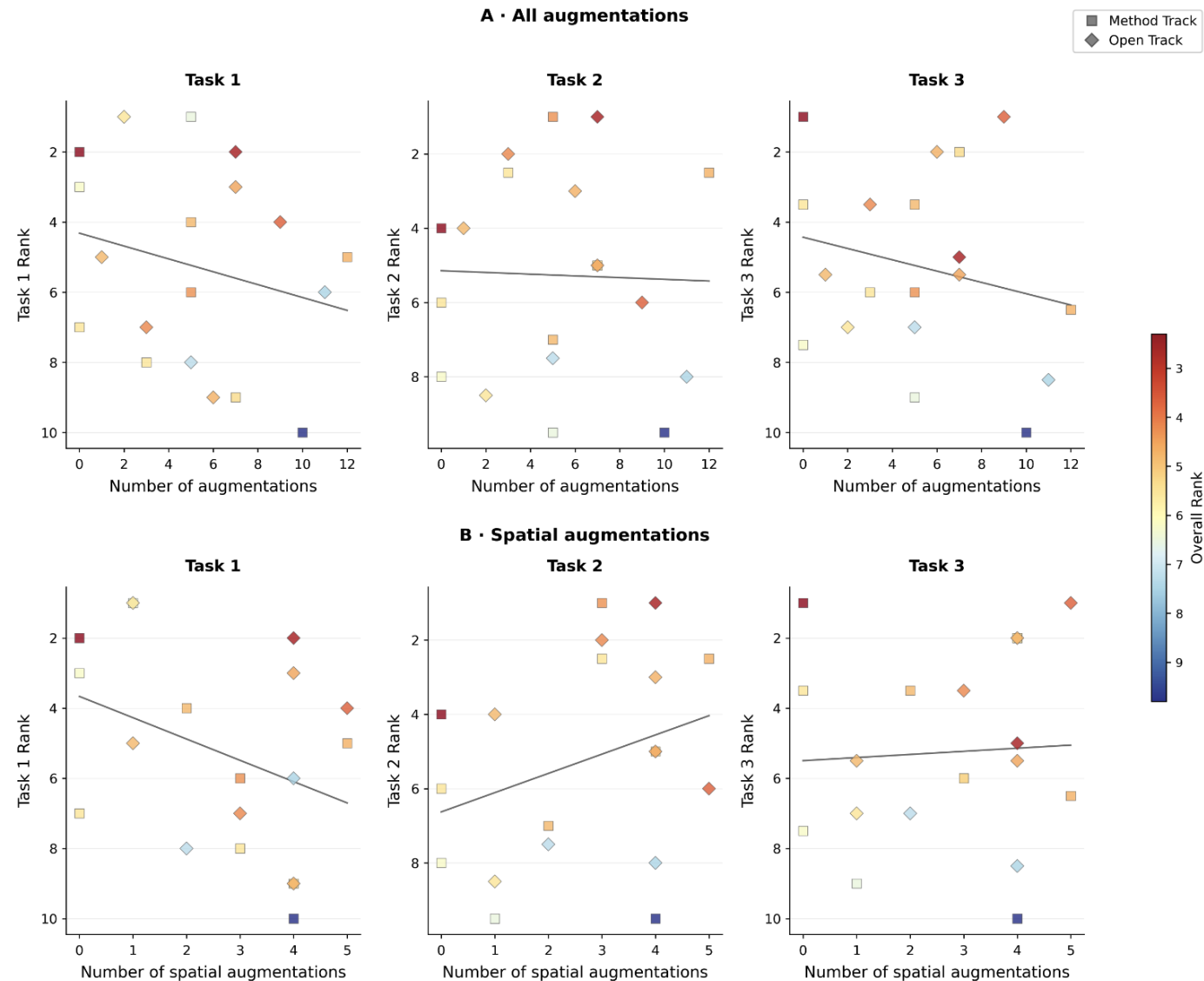


Figure 4: **Effect of augmentation strategies** on task performance. (A) Task-specific rank as a function of the total number of pretraining augmentations. (B) Task-specific rank as a function of the number of spatial augmentations only. Marker shapes denote track, and colors indicate overall rank within each track. Lower rank indicates better performance. Trend lines are shown for each task.

- Pretraining Boosts Generalization
 - Supervised baseline은 few-shot, OOD 조건에서 poor performance
 - Foundation model은 cross-continent domain gap을 few-shot으로 극복
- Diversity and Immaturity of Current Pretraining Approaches
 - 다양한 방법론이 해당 필드의 미성숙함을 보여줌
 - 모든 model이 task 간 rank variation 큼
 - CNN이 많이 사용되는 것은 최신 패러다임과 대비됨
- Implications for Future Research Directions
 - 어떤 submission도 task saturation에 도달하지 못함

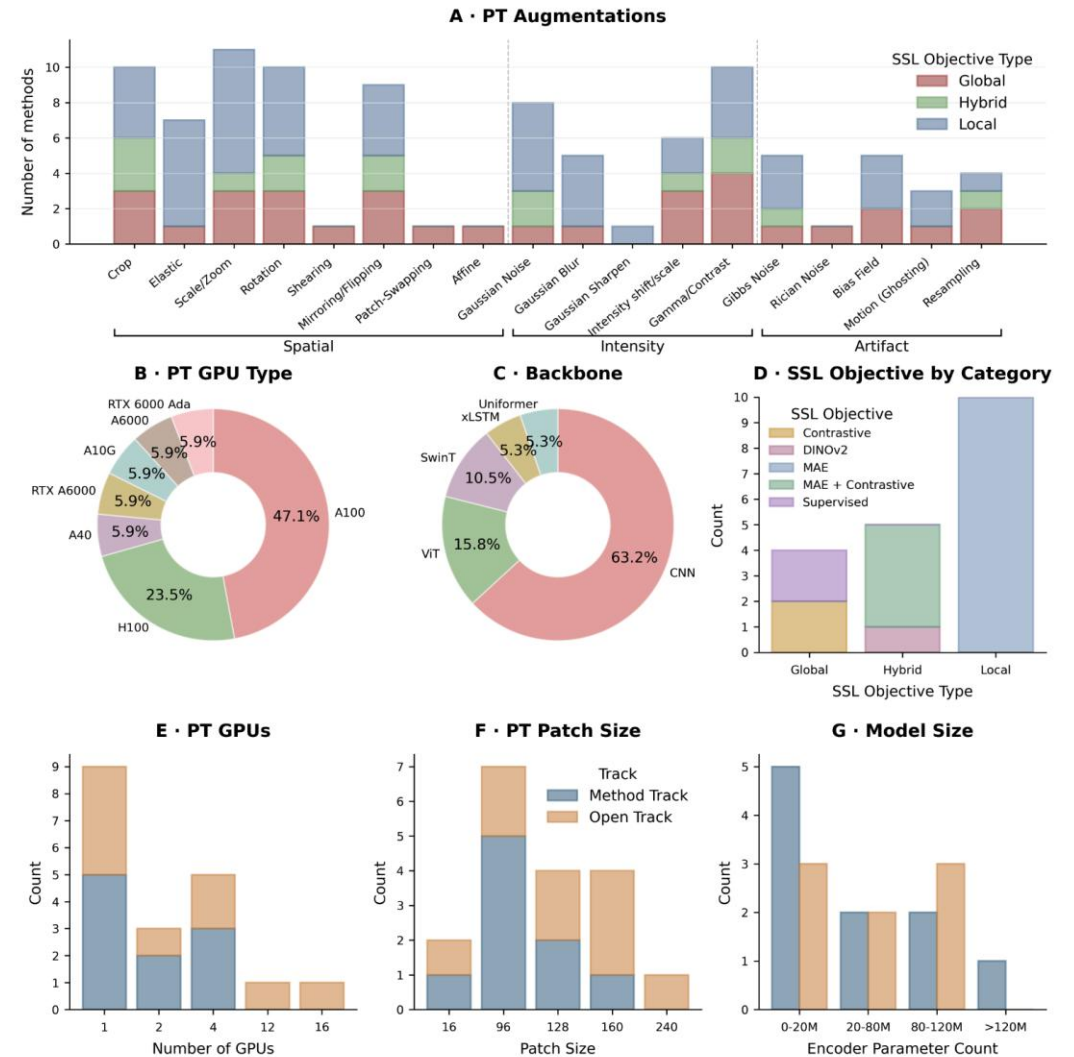


Figure B.9: **Overview of submitted methods.** (A) Pretraining augmentations used across all submissions, grouped by spatial, intensity, and artifact categories and colored by SSL objective type (Global, Hybrid, Local). (B) Distribution of GPU types used for pretraining. (C) Backbone architectures, showing the proportion of CNN, Swin Transformer, Uniformer, Vision Transformer, and xLSTM models. (D) Count of SSL objective types by category. (E) Number of GPUs used during pretraining. (F) Pretraining patch sizes, stratified by track. (G) Encoder parameter counts. Panels E–G separate submissions per track.

- FOMO25: 3개 task
 - **FOMO60K** - 11,187 subjects / 13,900 sessions / 60,529 scans
 - Infarct Classification (few-shot, AUROC)
 - Meningioma Segmentation (few-shot, DSC+NSD)
 - Brain Age Regression (MAE+correlation)
- FOMO26: 7개 task (Task 1-3은 동일, Task 4-7은 신규)
 - **FOMO300K - 58,741 subjects / 81,195 sessions / 306,207 scans**
 - Task 4 (신규): Trigeminal Neuralgia Segmentation — 삼차신경 + 주변 혈관 multiclass segmentation (수술 계획용, T2w)
 - Task 5 (신규): Polymicrogyria Classification — 희귀 피질 기형 few-shot 분류 (T1w)
 - Task 6 (신규): Linear Probing — finetuning 없이 embedding 품질만 평가 (pretraining에서 무엇을 학습했는지 직접 측정)
 - Task 7 (신규): Bias & Fairness — Task 6과 동일 setup으로 demographic/acquisition 편향 평가

Dataset	Scans	Sessions	Co-registered	Disease Metadata	Clinical-grade (scans)	Control	Brain Tumor	Stroke	Mental Disorders	Dementia	Neurological Disorders
OpenMind	114K	46K (35K)	✗	✗	8K (7%)	30K (85%)	0.2K (1%)	2.5K (7%)	1.2K (4%)	0 (0%)	0.7K (2%)
FOMO50K (ours)	49K	14K (9K)	✓	✓	5K (10%)	3K (32%)	2.7K (32%)	2.5K (28%)	0.1K (2%)	0.3K (3%)	0 (0%)
FOMO300K (ours)	306K	83K (67K)	✗	✓	53K (17%)	44K (65%)	16K (23%)	2.6K (4%)	2.1K (3%)	0.7K (1%)	0.9K (1%)

